

Integration — Lesson 3 Test: Slicing Curves into Rectangles

10 questions · show your work in the box · calculators allowed · every answer needs UNITS where they apply

Name: _____ Date: _____ Score: _____ / 10

1. Water flows into a tank at an increasing rate, measured once per minute: $t = 0$: 3 L/min, $t = 1$: 5, $t = 2$: 7, $t = 3$: 9, $t = 4$: 11. Estimate the total water from $t = 0$ to $t = 4$ with a LEFT sum (4 slices of width 1 minute). 1 pt

2. Using the same table as question 1, estimate the total water with a RIGHT sum, and state which of your two estimates is the overestimate. 1 pt

3. Compute the LEFT sum for the area under $y = 2x$ from $x = 0$ to $x = 3$ using 3 slices of width 1. (First find the heights at $x = 0, 1, 2$.) 1 pt

4. For the same setup as question 3 ($y = 2x$ from 0 to 3, three slices of width 1): compute the RIGHT sum, then compute the TRUE area using the triangle formula ($\frac{1}{2} \times \text{base} \times \text{height}$) and check that it lands between your left and right sums. 1 pt

5. You want to estimate an area from $t = 2$ to $t = 6$ using 8 equal slices. How wide is each slice? List the first three slice edges. 1 pt

6. Estimate the area under $y = x^2$ from $x = 0$ to $x = 4$ using 2 slices of width 2. Compute BOTH the left sum and the right sum. (Heights: $0^2 = 0$, $2^2 = 4$, $4^2 = 16$.) 1 pt

7. A drone's speed graph is INCREASING for the whole flight. Which estimate is bigger, the left sum or the right sum? Explain WHY in one or two sentences (talk about where each rectangle's top sits compared to the curve). 1 pt

8. A bathtub drains at a DECREASING rate: $t = 0$: 12 L/min, $t = 1$: 9, $t = 2$: 6, $t = 3$: 4. For 1 pt
the total water drained from $t = 0$ to $t = 3$ (slices of width 1), compute the left sum and the right sum, and label each one as an overestimate or an underestimate.

9. For an increasing rate graph, your left sum is 30 and your right sum is 38. A classmate 1 pt
claims the true total is 40. Explain how you know the classmate is wrong, and state what the true total COULD be.

10. For $y = 2x$ from $x = 0$ to $x = 4$ with 2 slices of width 2: compute the left sum, the right 1 pt
sum, and their AVERAGE. Then compute the true triangle area and say what you notice about the average.
